Name _____ **Surname** _____ **Date** ____/____/_____

Midterm test: Measurement

Duration: 70 minutes. **Total:** 100 points. A calculator is allowed. Write your answers on this sheet. Show your working in every calculation, and give final answers with the correct unit and the correct number of significant figures. A data sheet is on the last page.

Safety net. The questions marked ⛑ are worth 60 points out of 100, that is 60%: if you answer all of them correctly, you pass.

Exercise 1 — Quantities and units — 10 points

(a) ⛑ The SI base unit of mass is: **[2 p.]**
☐ A gram ☐ B kilogram ☐ C milligram ☐ D newton

(b) ⛑ Which of these is a *derived* quantity? **[2 p.]**
☐ A length ☐ B mass ☐ C time ☐ D density

(c) ⛑ The unit of density is: **[2 p.]**
☐ A m^3/kg ☐ B $kg\,m^3$ ☐ C kg/m^3 ☐ D N/m^2

(d) ⛑ 5 km is equal to: **[2 p.]**
☐ A 0.005 m ☐ B 50 m ☐ C 5000 m ☐ D 500 000 m

(e) ⛑ True or false? "38.4×10^4 m is written in standard form." **[2 p.]**
☐ True ☐ False

Exercise 2 — Significant figures — 10 points

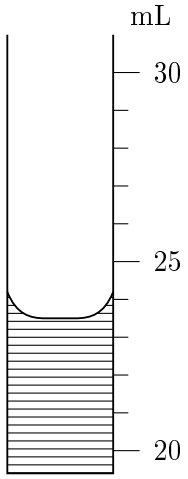
⛑ Write the number of significant figures of each measurement. **[2 p.]** each

(a) 0.0072 m _____ (b) 207 s _____ (c) 5.00 kg _____

(d) 0.03040 m _____ (e) 2.80×10^3 m _____

Exercise 3 — Reading instruments — 10 points

The figure shows part of the scale of a measuring cylinder. The water level is shown.



(a) ⛑ What is the value of one division of the scale? **[2 p.]**
Answer: _____

(b) ⛑ Read the volume of water, in mL. **[3 p.]**
Answer: _____

(c) ⛑ What is the uncertainty of one reading on this cylinder? **[2 p.]**
Answer: _____

(d) ⛑ A digital balance shows 94.0 g. What is the uncertainty of this reading? **[2 p.]**
Answer: _____

(e) ⛑ To read the level correctly, your eye should be: **[1 p.]**
☐ above the water ☐ level with the bottom of the curved surface ☐ below the water

Exercise 4 — Random or systematic error? — 10 points

For each situation, tick the type of error. **[2 p.]** each

	Random	Systematic
(a) A student starts a stopwatch sometimes a little early and sometimes a little late.	☐	☐
(b) A balance reads 0.2 g with nothing on the pan.	☐	☐
(c) A paper cone falls. The wind changes in a random way, so the cone sometimes falls faster and sometimes slower.	☐	☐
(d) A student always reads a measuring cylinder with her eye above the level of the water.	☐	☐

(e) Repeating a measurement and taking the mean reduces: **[2 p.]**

☐ random errors ☐ systematic errors ☐ both ☐ neither

Exercise 5 — True or false? — 10 points

Tick True or False. No explanation is needed. **[2 p.]** each

	True	False
(a) You press start and stop on a stopwatch with your finger. The stopwatch shows 0.01 s, so your time is correct to ± 0.01 s.	☐	☐
(b) If you start and stop a stopwatch with the same delay, the time you read is still correct.	☐	☐
(c) Taking the mean of many readings removes a zero error.	☐	☐
(d) The range of a set of readings is the largest value minus the smallest value.	☐	☐
(e) Readings that are close together but all far from the true value are precise but not accurate.	☐	☐

Exercise 6 — Conversions, standard form, rounding — 10 points

Answer each question. **[2 p.]** each

(a) Convert 450 mg to g. Answer: _____

(b) Convert 54 km/h to m/s. Answer: _____

(c) Write 7 200 000 m in standard form. Answer: _____

(d) Write 0.000036 s in standard form. Answer: _____

(e) Round 3.9871 kg to 2 significant figures. Answer: _____

Exercise 7 — A rolling ball — 10 points

A student times a ball rolling down a ramp six times with a stopwatch.

Reading	1	2	3	4	5	6
Time / s	1.62	1.58	1.64	1.60	1.55	1.61

(a) Calculate the mean time. Give your answer to 3 significant figures. **[3 p.]**

(b) Calculate the range of the readings. **[2 p.]**

(c) Suggest one reason why the readings are not all equal.

_____ **[2 p.]**

(d) The true time of the ball is 1.60 s. Another student starts the stopwatch 0.30 s late and stops it exactly on time. What time does the stopwatch show? _____ **[1 p.]**

(e) This student does the same every time. Is his error random or systematic? Explain in one sentence.

_____ **[2 p.]**

Exercise 8 — Density of a metal cube — 10 points

A student measures a metal cube. The side is $l = 3.0 \pm 0.1$ cm and the mass is $m = 213.0 \pm 0.1$ g. Use the rules on the data sheet.

(a) Calculate the relative uncertainty of l, in %. _____ **[1 p.]**

(b) Calculate the volume $V = l^3$ and its relative uncertainty. **[2 p.]**

$V =$ _____ relative uncertainty = _____

(c) Calculate the absolute uncertainty of V. _____ **[1 p.]**

(d) Calculate the relative uncertainty of m, in %. _____ **[1 p.]**

(e) Calculate the density $\rho = m/V$ and write the final result as $\rho \pm$ uncertainty, in g/cm^3. **[3 p.]**

(f) Which materials in the table on the data sheet are compatible with your result? Can you decide which metal the cube is made of? Explain in one sentence.

_____

_____ **[2 p.]**

Exercise 9 — Which container? **10 points**

A student wants the density of a small pebble of mass $m = 38.2 \pm 0.1$ g. She measures its volume by water displacement, using two different containers.

	Range	**One division**	V_1	V_2
Cylinder	100 mL	1 mL	50.0 mL	65.0 mL
Beaker	250 mL	25 mL	150 mL	175 mL

(a) Write the uncertainty of one reading for each container. **[2 p.]**

Cylinder: _____ Beaker: _____

(b) For each container calculate $V = V_2 - V_1$, its uncertainty and its relative uncertainty. **[3 p.]**

(c) Which container should she use? Explain in one sentence.

_____ **[1 p.]**

(d) Using the better container, calculate the density and write the final result as $\rho \pm$ uncertainty, in g/cm^3. **[3 p.]**

(e) A classmate says: "It must be granite, because 2.6 is in the table." Is he right? Use the rock table on the data sheet.

_____ **[1 p.]**

Exercise 10 — Two students, one swinging apple **10 points**

An apple swings on a string. The true time of one complete swing is 2.50 s. Two students time it five times with a stopwatch.

Student X (s)	2.80	2.78	2.83	2.79	2.81
Student Y (s)	2.35	2.62	2.48	2.71	2.40

(a) Calculate the mean of each student's readings, to 3 significant figures. **[2 p.]**

(b) Calculate the range of each student's readings. **[2 p.]**

(c) Which student has the larger random error? Which student has a systematic error, and how large is it?

[2 p.]

(d) Student X says: "If I take 100 more readings, my mean will get to 2.50 s." Is he right? Explain.

[2 p.]

(e) Suppose you did not know the true value. Could you tell from X's readings alone that X has a systematic error? Explain.

[2 p.]

Data sheet

Prefixes. 1 km $= 10^3$ m 1 mg $= 10^{-3}$ g 1 μs $= 10^{-6}$ s 1 nm $= 10^{-9}$ m

Standard form. $a \times 10^n$ with $1 \le a < 10$ and n an integer.

Other. 1 h = 3600 s 1 mL = 1 cm^3 density $\rho = \dfrac{m}{V}$

Rules of uncertainty

1. Analogue instrument: the uncertainty of one reading is half a division.
2. Digital instrument: the uncertainty of one reading is the last digit shown.
3. Difference of two readings ($V_2 - V_1$): the uncertainties add.
4. Relative uncertainty $= \dfrac{\text{uncertainty}}{\text{value}} \times 100\%$.
5. Product or quotient: the relative uncertainties add. For a cube, $V = l^3$, so the relative uncertainty of V is 3 times that of l.
6. Absolute uncertainty = relative uncertainty $\times$ value. Round the uncertainty to one significant figure.
7. Read a liquid level at the bottom of the curved surface, with your eye level with it.

Densities (g/cm^3, approximate)

Material	ρ	**Rock**	ρ
Water	1.00	Sandstone	2.2–2.8
Aluminium	2.7	Limestone	2.3–2.7
Steel	7.8	Granite	2.6–2.7
Brass	8.5	Marble	2.6–2.8
Copper	8.9	Slate	2.7–2.8
		Basalt	2.8–3.0